

8-BIT ML

Logistic Regression

Analyzing the *Titanic* dataset on vintage hardware: can we get a predictive model to converge?

ALGORITHM

- Run on ~200 points from Titanic Passenger Dataset (class, gender, age, deceased)
- Like a linear regression, but minimizes log-loss instead of sum of distances to line
- Fit by gradient descent



age	gender	fare	surv?
47	F	180	Y
34	M	220	N
27	F	95	N
56	M	310	?

DEMO

- Left-hand panel shows 2 features (x: gender, y: age)
- As the fit converges, the separation curve will be plotted on the left as well
- Right-hand panel shows loss history

IMPLEMENTATION

- Extended Graphics BASIC 4 on TRS-80 Model 4P (~1983)
- CPU: Z80A @ 4 MHz
- RAM: 128 kB

